Diffusion Limited Aggregation

Candidate Number: [Your Number]

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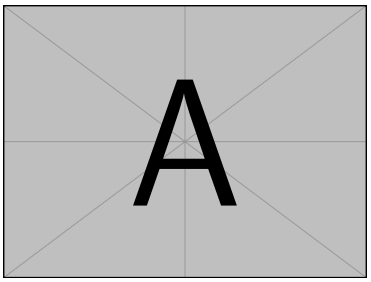


A dissertation submitted for the degree of
Mathematics Part B

Abstract

This dissertation investigates the universality class of Diffusion Limited Aggregation (DLA) through the lens of conformal mapping and harmonic measure theory. We compare on-lattice, off-lattice, and hybrid "snapping" algorithms to determine if microscopic lattice anisotropy survives in the macroscopic scaling limit. Particular attention is paid to the dimension of the harmonic measure and the dynamics of the active growth zone. [Placeholder: Finish this summary with your specific results regarding the Hybrid model's classification, totaling approx. 300 words.]

Contents

1 Introduction **4**

1.1 Overview . . . 4

1.2 Theoretical Context . . . 4

1.3 Fractal Dimension . . . 5

1.4 Anisotropy and Thesis Goals . . . 5

2 Potential Theory **6**

2.1 Preliminaries . . . 6

2.1.1 Martingales and the Optional Stopping Theorem . . . 6

2.1.2 Measure Theory Concepts . . . 7

2.1.3 Conformal Maps . . . 8

2.2 Discrete Potential Theory . . . 8

2.2.1 Simple Random Walks and the Generator . . . 8

2.2.2 The Discrete Dirichlet Problem . . . 9

2.2.3 Recurrence and the Potential Kernel . . . 11

2.2.4 Discrete Harmonic Measure . . . 12

2.3 Continuous Potential Theory . . . 14

2.3.1 Brownian Motion and the Laplacian . . . 14

2.3.2 Harmonic Measure and the Poisson Kernel . . . 16

2.3.3 Conformal Invariance . . . 17

2.3.4 Solvable Geometries . . . 17

3 Algorithms **20**

3.1 Introduction . . . 20

3.2 Initialisation . . . 20

3.2.1 Off-Lattice: The Launching Circle . . . 20

3.2.2 On-Lattice: The Fuzzy Annulus . . . 21

3.3 Far-Field Diffusion . . . 21

3.3.1 Off-Lattice: Re-injection from Infinity . . . 21

3.3.2 On-Lattice: Walk-to-Line . . . 22

3.4 Near-Field Diffusion . . . 23

3.4.1 Off-Lattice: Walk-to-Circle . . . 23

3.4.2 On-Lattice: Walk-to-Square . . . 24

3.5 Aggregation . . . 24

3.5.1 On-Lattice: Sticky Boundary . . . 24
3.5.2 Off-Lattice: Exact Adhering . . . 24
3.6 The Hybrid Model . . . 26
3.6.1 Motivation and Design . . . 26
3.6.2 Implementation . . . 27

A Implementation Details and Optimisations 30
A.1 Computational Framework . . . 30
A.1.1 The Numba JIT Compiler . . . 30
A.1.2 Rounding Semantics . . . 30
A.2 Spatial Indexing and Hierarchical Skipping . . . 31
A.2.1 Lattice: Flattened Bit-Map . . . 31
A.2.2 Off-Lattice: Dual-Structure Strategy . . . 32
A.2.3 2. The Spatial Hash (Hitting) . . . 32
A.3 Rejection Sampling . . . 32
A.3.1 Concept and Motivation . . . 32
A.3.2 The Algorithm . . . 33
A.3.3 Application 1: The Fuzzy Annulus . . . 33
A.3.4 Application 2: Walk-to-Line . . . 33
A.4 Walker's Alias Method . . . 34
A.4.1 Motivation . . . 34
A.4.2 Implementation . . . 34
A.5 Numerical Approximations . . . 35
A.5.1 Custom Bessel Functions . . . 35
A.6 Usability and Data Management . . . 35
A.6.1 Heuristic Grid Sizing . . . 35
A.6.2 Standardised Coordinate Output . . . 35

B Reference: LaTeX Structure Examples 37
B.1 Aligned Equations . . . 37
B.2 Floats . . . 37
B.2.1 Single Figure . . . 37
B.2.2 Side-by-Side Figures (Subfloats) . . . 37
B.3 Algorithms . . . 39
B.4 Theorems . . . 39
B.5 References . . . 40

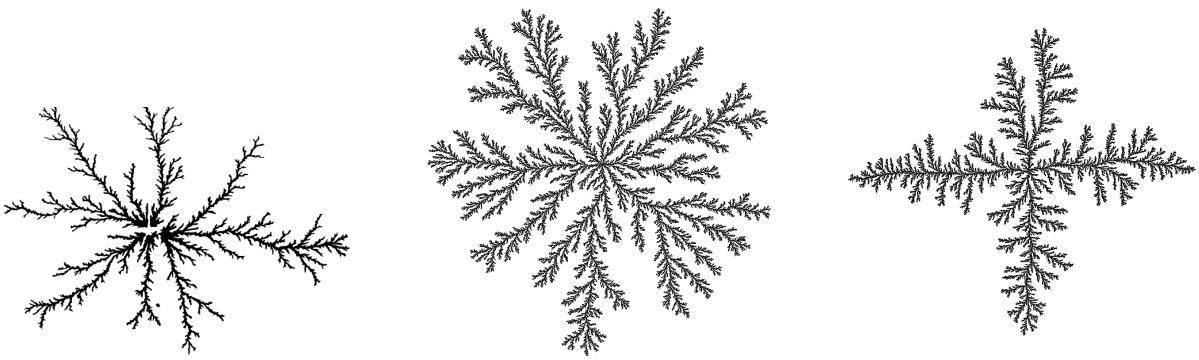
Chapter 1

Introduction

1.1 Overview

Diffusion Limited Aggregation (DLA) is a stochastic model for irreversible colloidal aggregation. In this model a seed particle is placed at the origin, and then subsequent particles are released from a boundary far from the cluster one at a time. The particle then diffuses, either as a random walk on a lattice or Brownian motion (the continuous analogue to a random walk). Upon collision, the walker sticks and the next one starts diffusing.

The model was first proposed in 1981 by Witten and Sander to model the colloidal aggregation of metal precipitates [1], but has since been found to have applications in a diverse range of physical systems, including electrodeposition [2], bacteria growth [3] and viscous fingering in Hele-Shaw cells [4] among others.



(a) Zinc Electrodeposition [2] **(b)** Off-Lattice DLA **(c)** On-Lattice DLA

Figure 1.1: Comparison of physical aggregation (a) with the two primary modelling approaches: continuum Off-Lattice DLA (b) and discrete On-Lattice DLA (c). Both are the results of simulation with 10^7 particles. Note the distinct grid-induced anisotropy visible in the on-lattice case.

1.2 Theoretical Context

While DLA is defined by a stochastic algorithm (random walkers), its theoretical significance lies in its connection to potential theory, the study of harmonic functions. As will be detailed in the following chapters, the probability density $\phi(u)$ of a diffusing particle at position u satisfies the Laplace equation $\nabla^2\phi = 0$. Intuitively, this is because both random walks and harmonic

functions are defined by the property that the value at any point is the average of its neighbours. Each additional walker to aggregate onto the cluster distorts this probability field, creating a Laplacian instability that amplifies irregularities into the observed complicated structure [5].

1.3 Fractal Dimension

Visually, DLA clusters exhibit self-similarity, where small branches look statistically identical to large branches. Mathematically this can be understood via fractal dimension D. This will be formally defined later, but can be thought of as the relationship between the mass of a cluster (number of particles N) and its linear size R, where:

$$N \propto R^D. \tag{1.1}$$

For instance, doubling the size of a line would double its mass, so it has dimension 1, while doubling the size of a solid disc would quadruple its mass, so this has dimension 2. Simulations of off-lattice DLA have consistently yielded a fractal dimension $D \approx 1.71$, but this has not been proven [6].

In fact, the only mathematical theorem on 2D DLA is due to Kesten, who proved a bound on the growth rate of on-lattice DLA in 1987, with a key consequence being that the fractal dimension of a DLA cluster is bounded below by 1.5 [7].

1.4 Anisotropy and Thesis Goals

It is likely that the dimension of on-lattice DLA is actually lower than off-lattice due to the effect of the grid on the cluster shape, a phenomenon known as anisotropy. Grebenkov and Beliaev discussed how anisotropy is likely to reduce the dimension of large enough clusters [8], but understanding why and to what extent is still not fully understood.

To answer these questions, simulations of DLA must be done both on and off-lattice. Additionally, to understand the effect of the lattice we create a 'hybrid' algorithm with continuous diffusion but a discrete aggregation method. Other hybrid models of DLA have previously been studied [9], including a similar semi-lattice approach [10], but to our knowledge have not been utilised to decouple the effects of transport from aggregation in on-lattice DLA.

The purpose of this dissertation is to compare these three different models of DLA, in order to decouple the diffusion process of DLA from the geometry of the aggregation. By analysing various features of the resulting clusters, we hope to better understand the relationship between the continuous and discrete variations of the probability distribution governing DLA growth.

We begin with a brief outline of the mathematical necessary to understand the underlying mechanisms behind DLA in chapter 2, before following on by explaining how to utilise this theory to create efficient algorithms, as well as discussing our hybrid approach, which forms chapter 3. In ?? we begin analysis, focusing on ???

Chapter 2

Potential Theory

2.1 Preliminaries

2.1.1 Martingales and the Optional Stopping Theorem

In this section, we review the probabilistic tools required to link random walks to potential theory. We assume standard measure-theoretic knowledge (equivalent to the Part A Integration course).

Definition 2.1 (Probability Space)**.** *A probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a triple where* Ω *is a sample set,* $\mathcal{F}$ *is a* σ*-algebra on* Ω *representing the set of events, and* $\mathbb{P}\colon \mathcal{F} \to [0,1]$ *is a countably additive set function such that* $\mathbb{P}(\Omega) = 1$*.*

Definition 2.2 (Filtration)**.** *A filtration on a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a sequence of sub-*σ*-algebras* $(\mathcal{F}_n)_{n\geq 0}$ *such that:*

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

The tuple $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\geq 0}, \mathbb{P})$ *is called a filtered probability space.*

Intuition: The filtration $(\mathcal{F}_n)$ represents the accumulation of information over time. $\mathcal{F}_n$ contains all events whose occurrence (or non-occurrence) is known by time n.

Definition 2.3 (Stopping Time)**.** *A random variable* $\tau\colon \Omega \to \{0, 1, \ldots, \infty\}$ *is called a stopping time with respect to the filtration* $(\mathcal{F}_n)_{n\geq 0}$ *if for every* $n \geq 0$*:*

$$\{\omega \in \Omega : \tau(\omega) \leq n\} \in \mathcal{F}_n.$$

Intuition: A stopping time represents a strategy where the decision to stop at time n depends only on the history of the process up to time n. For example, "stop when the walker hits the cluster" is a valid stopping time but "stop at the step before the walker hits the cluster" is not.

Definition 2.4 (Martingale)**.** *A discrete-time stochastic process* $(X_n)_{n\geq 0}$ *is a martingale with respect to a filtration* $(\mathcal{F}_n)_{n\geq 0}$ *if:*

1. X_n *is adapted to the filtration (i.e.,* X_n *is* $\mathcal{F}_n$*-measurable for all* n*),*
2. $\mathbb{E}[|X_n|] < \infty$ *for all* n *(integrability),*
3. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ *almost surely (conditional expectation).*

Intuition: A martingale models a “fair game" with no drift. The condition $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ implies that given the entire history of the process, the best prediction for the next value is the current value.

Theorem 2.1 (Doob's Optional Stopping Theorem)**.** *Let X be a martingale on a filtered probability space. If τ is a stopping time such that $\mathbb{P}(\tau < \infty) = 1$, and X is uniformly bounded (i.e., there exists K such that $|X_n(\omega)| \leq K$ for all n, ω), then:*

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0]. \tag{2.1}$$

Proof. See Williams [11, Section 10.9]. □

Intuition: The Optional Stopping Theorem states that for a fair game, no stopping strategy τ can change the expected payout. This will allow us to relate the value of a harmonic function at a starting point x (start of the walk) to its expected value at the boundary ∂D (end of the walk).

2.1.2 Measure Theory Concepts

In the study of potential theory, we will want to understand the relationship between the geometric length of a boundary (Lebesgue measure) and the probability that a random walker hits that boundary (harmonic measure). To rigorously define a probability density function linking these two concepts we require the notion of the Radon-Nikodym derivative.
Remark: throughout this work, we assume our measurable spaces are equipped with the Borel σ-algebra.

Definition 2.5 (Absolute Continuity)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space equipped with two measures μ and ν. We say that ν is absolutely continuous with respect to μ, denoted $\nu \ll \mu$, if for every $A \in \mathcal{F}$:*

$$\mu(A) = 0 \implies \nu(A) = 0. \tag{2.2}$$

Intuition: If $\nu \ll \mu$, then ν assigns zero mass to any set where μ assigns zero mass. This ensures that we can't have a "probability" of being in a region with zero "volume."

Theorem 2.2 (The Radon-Nikodym Theorem)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space, and let μ and ν be two finite measures such that $\nu \ll \mu$. Then, there exists a non-negative measurable function $f \colon \Omega \to [0, \infty)$, unique μ-almost everywhere, such that for any $A \in \mathcal{F}$:*

$$\nu(A) = \int_A f \, d\mu. \tag{2.3}$$

The function f is called the Radon-Nikodym derivative of ν with respect to μ and is denoted by:

$$f = \frac{d\nu}{d\mu}. \tag{2.4}$$

Proof. See Williams [11, Section 5.14]. □

2.1.3 Conformal Maps

Here we state the definition of a conformal map, and the Riemann mapping theorem. Conformal maps will be very important for two reasons: theoretically, the Riemann Mapping Theorem guarantees that the Dirichlet problem is well-posed even on the fractal boundary of a DLA cluster; algorithmically, these maps allow us to exploit the 'conformal invariance' of Brownian motion (discussed later)

Definition 2.6 (Conformal Map)**.** *Let $U, V \subseteq \mathbb{C}$ be open sets. A function $f\colon U \to V$ is called* ***conformal*** *at a point $z_0 \in U$ if it preserves angles and orientation locally.*

Analytically, a holomorphic function f is conformal at every point where its derivative is non-zero, i.e., $f'(z) \neq 0$. If f is a holomorphic bijection between U and V (a biholomorphism), it is essentially a global conformal map, and we say U and V are conformally equivalent.

Theorem 2.3 (Riemann Mapping Theorem)**.** *Let $U \subsetneq \mathbb{C}$ be a non-empty, simply connected open subset of the complex plane (i.e., U is the whole plane minus at least one point, with no holes). Then, there exists a unique bijective conformal map f from U onto the open unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ such that:*

$$f(z_0) = 0 \quad \textit{and} \quad f'(z_0) > 0, \tag{2.5}$$

for any prescribed point $z_0 \in U$.

Proof. See Ahlfors [12, Chapter 6] or Priestley [13, Chapter 9]. □

2.2 Discrete Potential Theory[1]

We now introduce potential theory in a discrete setting, focusing on its applications to DLA. We start by considering the definitions of a random walk and a discrete harmonic function.

2.2.1 Simple Random Walks and the Generator

Definition 2.7 (Random Walk)**.** *A random walk on $\mathbb{Z}^2$ is a stochastic process $(S_n)_{n\geq 0}$ defined by $S_n = S_0 + \sum_{i=1}^{n} X_i$, where $X_1, X_2, \ldots$ are independent and identically distributed random variables taking values in $\mathbb{Z}^2$.*

The common distribution of the steps is denoted by $p(x) = \mathbb{P}(X_i = x)$ and is called the ***increment distribution****.*

For the purposes of DLA, we are primarily interested in the symmetric case:

Definition 2.8 (Simple Symmetric Random Walk)**.** *A random walk is called a Simple Symmetric Random Walk (SSRW) if the increment distribution is uniform on the nearest neighbours:*

$$p(x) = \begin{cases} \frac{1}{4} & \textit{if } |x| = 1, \\ 0 & \textit{otherwise.} \end{cases} \tag{2.6}$$

[1] The theoretical framework in this section is largely adapted from Lawler and Limic [5], applied here to the specific case of 2D DLA.

We can now define the operator that governs the evolution of the expectation of functions under this flow.

Definition 2.9 (Discrete Generator)**.** *Let $p(\cdot)$ be the increment distribution of a random walk. The generator $\mathcal{L}$ is a linear operator acting on functions $f : \mathbb{Z}^2 \to \mathbb{R}$ defined by:*

$$\mathcal{L}f(x) = \sum_{y \in \mathbb{Z}^2} p(y - x)[f(y) - f(x)]. \tag{2.7}$$

Probabilistically, this represents the expected change in the function f in a single step of the walk starting at x:

$$\mathcal{L}f(x) = \mathbb{E}[f(S_1) - f(S_0) \mid S_0 = x]. \tag{2.8}$$

Remark: For Simple Symmetric Random Walk, the generator is proportional to the unnormalised discrete Laplacian, $\mathcal{L} = \frac{1}{4}\Delta$, where $\Delta f(x) = \sum_{y \sim x}(f(y) - f(x))$.

Definition 2.10 (Discrete Harmonic Functions)**.** *A function $f : A \cup \partial A \to \mathbb{R}$ is said to be harmonic in a subset $A \subset \mathbb{Z}^2$ if:*

$$\mathcal{L}f(x) = 0 \quad \textit{for all } x \in A. \tag{2.9}$$

Proposition 2.1 (Harmonic Functions are Martingales)**.** *Let S_n be a random walk and f be a function harmonic on a set A. Let $\tau_A = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time from A. Then the stopped process $M_n = f(S_{n \wedge \tau_A})$ is a martingale (with respect to the natural filtration of S_n).*

Proof. See Lawler and Limic [5, Proposition 6.1.1]. □

2.2.2 The Discrete Dirichlet Problem

Definition 2.11 (Boundary)**.** *Before proceeding, we distinguish between the occupied cluster and the available growth sites. Let $K \subset \mathbb{Z}^2$ denote the set of occupied sites (the cluster). We define the boundary ∂K as the outer boundary consisting of all empty sites adjacent to the cluster:*

$$\partial K = \{y \in \mathbb{Z}^2 \setminus K : \exists x \in K \textit{ such that } |y - x| = 1\}. \tag{2.10}$$

Subsequently we refer to the outer boundary as the ***boundary*** *of the cluster ∂K.*

We now attempt to model DLA as a discrete Dirichlet problem, where the unknown function $u_R(x)$ is defined as the probability of a random walker starting at x escaping to some outer boundary ∂R of radius R before hitting the cluster boundary ∂K.

$$u_R(x) = \mathbb{P}^x(\text{hit outer boundary } \partial R \text{ before } \partial K). \tag{2.11}$$

As the diffusion process is a simple random walk, the probability of escaping from x is simply the average of the excape probabilities of its neighbours:

$$u_R(x) = \frac{1}{4} \sum_{z \sim x} u_R(z). \tag{2.12}$$

By definition this makes $u(\cdot)$ harmonic, hence satisfies the discrete Laplace equation $\mathcal{L}u(x) = 0$.

Thus, each stage of DLA is equivalent to solving the Dirichlet problem:

$$\begin{cases} \Delta f(x) = 0 & \text{if } x \notin K \text{ (Harmonic in free space)}, \\ f(x) = 0 & \text{if } x \in \partial K \text{ (Hit the cluster)}, \\ f(x) = 1 & \text{if } x \in \partial R \text{ (Escaped to infinity)}. \end{cases} \tag{2.13}$$

Remark 2.1. *We could have solved for the probability of hitting the cluster before escaping, reversing the boundary conditions. However, we do it this way to preserve consistency with the later sections and the electrostatic interpretation (see Appendix).*

We now look at the general discrete Dirichlet problem in 2D:

Theorem 2.4 (Solution to the 2D Discrete Dirichlet Problem)**.** *Let $A \subset \mathbb{Z}^2$ be a subset (potentially infinite) and let $\tau = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time. Assume that:*

1. *The random walk stops almost surely: $\mathbb{P}_x(\tau < \infty) = 1$ for all $x \in A$.*
2. *The boundary function $g : \partial A \to \mathbb{R}$ is bounded.*

Then, the function $f : A \cup \partial A \to \mathbb{R}$ defined by the expectation:

$$f(x) = \mathbb{E}_x[g(S_\tau)] \tag{2.14}$$

is the unique bounded solution to the Dirichlet system:

$$\mathcal{L}f(x) = 0, \qquad \text{for all } x \in A, \tag{2.15a}$$

$$f(x) = g(x), \qquad \text{for all } x \in \partial A. \tag{2.15b}$$

Proof. **Existence:** We first check that (2.14) satisfies the system (2.15). If $x \in \partial A$, then $\tau = 0$, so $S_\tau = x$ and $f(x) = \mathbb{E}_x[g(x)] = g(x)$. Thus the boundary condition (2.15b) holds. If $x \in A$, we use the Markov property at the first step:

$$f(x) = \mathbb{E}_x[g(S_\tau)] = \sum_{y \sim x} p(y - x)\mathbb{E}_y[g(S_\tau)] = \sum_{y \sim x} p(y - x)f(y). \tag{2.16}$$

Thus $\mathcal{L}f(x) = 0$, satisfying (2.15a).

Uniqueness: Suppose h is another bounded solution to the system (2.15). Since h satisfies (2.15a), it is harmonic, and by proposition 2.1, the process $M_n = h(S_{n\wedge\tau})$ constitutes a martingale. Since h is bounded, M_n is a bounded martingale.
Given that $\tau < \infty$ almost surely, we can apply the Optional Stopping Theorem (theorem 2.1):

$$h(x) = \mathbb{E}_x[M_0] = \mathbb{E}_x[M_\tau] = \mathbb{E}_x[h(S_\tau)]. \tag{2.17}$$

From condition (2.15b), we know $h(S_\tau) = g(S_\tau)$ on the boundary. Therefore:

$$h(x) = \mathbb{E}_x[g(S_\tau)], \tag{2.18}$$

which is exactly $f(x)$. Thus $h \equiv f$, proving uniqueness. □

2.2.3 Recurrence and the Potential Kernel

To apply this solution to DLA, we require taking the limit $R \to \infty$, which is equivalent to solving the Dirichlet problem with a boundary condition at infinity.

This is done formally by Lawler and Limic [5, Proposition 6.2.2], who prove that a solution to the Dirichlet problem in d dimensions is uniquely determined by values on the finite boundary ∂A and the value at infinity, taking the form:

$$f(x) = \mathbb{E}_x[g(S_\tau)\mathbf{1}_{\tau<\infty}] + g(\infty)\mathbb{P}_x(\tau = \infty). \tag{2.19}$$

However, due to the recurrence of a random walk in 2D space, the escape probability $\mathbb{P}_x(\tau = \infty)$ is zero for any random walk in $\mathbb{Z}^2$. Consequently, the second term vanishes. The solution is now entirely determined by the inner boundary ∂K. For our boundary conditions given in eq. (2.15), the only solution is the trivial potential $f \equiv 0$.

To address this we now turn to the concept of Green's functions. The Green's function $G(x, y)$ represents the potential field at x induced by a point source at y. Put another way, it is the solution to the Poisson equation:

$$\Delta G(x, y) = -\delta_y(x). \tag{2.20}$$

In the probabilistic setting, the Green's function has a precise interpretation: it is the expected number of times a random walk starting at x visits the site y:

$$G(x, y) = \mathbb{E}_x\left[\sum_{n=0}^{\infty} \mathbf{1}_{\{S_n=y\}}\right] = \sum_{n=0}^{\infty} p_n(y - x). \tag{2.21}$$

For transient random walks (such as in dimensions $d \geq 3$), this sum converges, and $G(x, y)$ decays to zero as $|x - y| \to \infty$.

However, in two dimensions the random walk is recurrent, meaning a walker returns to the origin infinitely often, causing the sum in the definition of $G(x, y)$ to diverge:

$$G(x, y) = \infty \quad \text{for all } x, y. \tag{2.22}$$

Instead we regularise the divergent Green's function by working with potential differences, motivating the definition of the potential kernel.

Definition 2.12 (Potential Kernel)**.** *The potential kernel $a(x)$ for simple random walk in $\mathbb{Z}^2$ is defined as:*

$$a(x) = \sum_{n=0}^{\infty}[p_n(0) - p_n(x)], \tag{2.23}$$

where $p_n(x) = \mathbb{P}(S_n = x)$.

It is not immediately clear that this will be finite. However, using estimates from the Local Central Limit Theorem [5, Section 4.4.1], it can be shown that for any fixed x, the difference decays as:

$$|p_n(0) - p_n(x)| \leq C|x|n^{-3/2}. \tag{2.24}$$

Since the series $\sum_{n=1}^{\infty} n^{-3/2}$ converges absolutely, the potential kernel $a(x)$ is well-defined and

finite for all x.

By transitioning from the Dirichlet problem to the potential kernel, we stop solving directly for the probability and instead solve for a renormalised potential $a(x)$. Intuitively, $a(x)$ quantifies the relative accessibility of a given lattice site from infinity. Importantly, the growth probability is recovered through the discrete differences of this potential: The proportion of walkers hitting a boundary point depends on the 'steepness' of the potential at the boundary.
The potential kernel therefore acts as the fundamental solution to the Laplacian on the lattice. It satisfies the Poisson equation:

$$\mathcal{L}a(x) = \delta_0(x) = \begin{cases} 1 & x = 0, \\ 0 & x \neq 0. \end{cases} \tag{2.25}$$

It replaces the 'boundary at infinity' with an asymptotic growth condition.

Theorem 2.5 (Asymptotics of the Potential Kernel)**.** *As $|x| \to \infty$, the potential kernel satisfies:*

$$a(x) = \frac{2}{\pi} \ln |x| + k + O(|x|^{-2}), \tag{2.26}$$

where k is a constant involving Euler's constant γ.

Proof. See Lawler and Limic [5, Theorem 4.4.4]. □

Remark 2.2. *The logarithmic growth may seem arbitrary, but can in fact be justified with some geometric intuition: Consider a large circle of radius R centred on the cluster (large so that it contains the entire cluster). For a steady flow of random walkers coming from infinity to reach the cluster, the total flux (rate of flow) across this circle must be independent of R. The total flux is the product of the circumference ($2\pi R$) and the radial gradient of the potential ($\nabla f(R)$). This implies that the gradient must decay as $\nabla f(R) \propto 1/R$. Integrating this gradient yields:*

$$f(R) \propto \int \frac{1}{R}\, dR = \ln R. \tag{2.27}$$

Non-logarithmic growth would imply that the flux across a circle varies depending on its radius, violating physical conservation laws.

2.2.4 Discrete Harmonic Measure

We now formally define the discrete harmonic measure:

Definition 2.13 (Harmonic Measure from Infinity)**.** *Let $K \subset \mathbb{Z}^2$ be a finite cluster. The harmonic measure $H_{\partial K}(y)$ is the probability that a simple random walk starting from infinity first hits the cluster K at the point $y \in \partial K$. Formally, it is defined as the limit:*

$$H_{\partial K}(y) = \lim_{|x| \to \infty} \mathbb{P}_x(S_{\tau_K} = y), \tag{2.28}$$

where $\tau_K = \inf\{n \geq 0 : S_n \in K\}$ is the first hitting time.

Lawler and Limic [5, Proposition 6.6.1] prove that this limit exists and defines a unique probability measure on the boundary ∂K. While the definition may seem abstract, we can in fact calculate these probabilities directly from the potential field derived in the previous section.

We begin by formally defining the concept of discrete flux which was invoked in remark 2.2. On the lattice, the concept of a normal derivative $\nabla u \cdot \mathbf{n}$ is replaced by a difference operator across the boundary.

Definition 2.14 (Discrete Flux)**.** *For a function $u : \mathbb{Z}^2 \to \mathbb{R}$ that vanishes on K, the discrete flux at a boundary point $y \in \partial K$ is the sum of the potential differences from its neighbours in the complement:*

$$\Phi(y) = \sum_{z \sim y, z \notin K} (u(z) - u(y)). \tag{2.29}$$

Since $u(y) = 0$ on the boundary, this simplifies to the sum of the external neighbours: $\Phi(y) = \sum_{z \sim y, z \notin K} u(z)$.

The fundamental result that we require from discrete potential theory is the equivalence of flux and the harmonic measure.

Proposition 2.2 (Harmonic Measure as Flux)**.** *The harmonic measure at $y \in \partial K$ is proportional to the discrete flux at y:*

$$H_{\partial K}(y) = \frac{\Phi(y)}{\sum_{w \in \partial K} \Phi(w)}. \tag{2.30}$$

Proof. For a formal rigorous proof connecting hitting probabilities to the Laplacian of the Green's function, see Lawler and Limic [5, Proposition 1.6.3]. □

To understand this link intuitively, we interpret the potential $u(x)$ as the density of a stream of random walkers originating from infinity and disappearing upon hitting the cluster K. In the exterior free space, the condition $\mathcal{L}u(x) = 0$ implies a sort of conservation law: the expected number of walkers entering any site x equals the number leaving it. However, at a boundary site $y \in \partial K$, walkers arrive from neighbours but do not continue; they form the cluster. The operator $\mathcal{L}u(y)$ represents this absorption rate:

$$\text{Rate into } y \propto \sum_{z \sim y} (\text{Density at } z) = \sum_{z \sim y} u(z) = \Phi(y). \tag{2.31}$$

Since the harmonic measure $H(y)$ is the proportion of all walkers that terminate at y, it is directly proportional to this local absorption rate, i.e., the flux $\Phi(y)$.

This concludes the discrete formulation of the DLA process. The growth mechanism can be summarised as follows:

1. The system is governed by a harmonic potential $u(x)$ determined by the geometry of the cluster K.

2. This potential is the solution to the discrete Laplace equation $\mathcal{L}u = 0$ with $u|_K = 0$ and $u(x) \sim \ln |x|$ as $|x| \to \infty$.

3. The growth probability at a boundary site y is determined by the local flux $\Phi(y) = \sum_{z \sim y} u(z)$, where a new particle is added with probability $H(y) \propto \Phi(y)$.

2.3 Continuous Potential Theory

Much of this section parallels the discrete theory presented in section 2.2, though the derivations here require the delicate machinery of stochastic calculus. However, for our purposes, an understanding of the main results is sufficient to build a robust intuition for DLA. The fundamental shift is replacing the discrete random walker with its scaling limit: Brownian motion.

2.3.1 Brownian Motion and the Laplacian

We start by defining Brownian motion in both $\mathbb{R}^d$ and $\mathbb{C}$.

Definition 2.15 (Standard Brownian Motion)**.** *A standard Brownian motion in $\mathbb{R}^d$ is a stochastic process $B = (B_t)_{t\geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, such that:*

1. *$B_0 = 0$ almost surely.*
2. *The process has independent increments: for any $0 \leq s < t$, $B_t - B_s$ is independent of $\mathcal{F}_s = \sigma(B_u : u \leq s)$.*
3. *The increments are Gaussian: $B_t - B_s \sim \mathcal{N}(0, (t-s)I_d)$.*
4. *The paths $t \mapsto B_t$ are continuous almost surely.*

In the study of 2D DLA, we will utilise results from complex analysis. Thus, it is natural to identify the plane $\mathbb{R}^2$ with $\mathbb{C}$.

Definition 2.16 (Complex Brownian Motion)**.** *A canonical complex Brownian motion Z_t is a process taking values in $\mathbb{C}$, defined as:*

$$Z_t = X_t + iY_t, \tag{2.32}$$

where X_t and Y_t are two independent, standard one-dimensional Brownian motions.

The utility of this complex formulation becomes clear when we consider what is known as the quadratic variation of the process. In stochastic calculus, standard Brownian motions satisfy the formal rules $(dX_t)^2 = dt$ and $(dY_t)^2 = dt$. Applying this yields:

$$(dZ_t)^2 = (dX_t + idY_t)^2 = (dX_t)^2 - (dY_t)^2 + 2idX_tdY_t. \tag{2.33}$$

Since X and Y are independent identical copies, their variances match ($dt - dt = 0$), and the cross-term is negligible for variation. Thus, we effectively have $(dZ_t)^2 = 0$. This property identifies Z_t as a conformal martingale, which is the stochastic equivalent of an analytic function (where derivative depends only on z, not $\bar{z}$, as given by Wirtinger derivatives), and will be why we get the conformal invariance result later.

Remark 2.3. *Much of this section has been somewhat hand-wavy. However, for our purposes a complete rigourous introduction to stochastic calculus is unnecessary. If nothing else, think of Brownian motion as the continuous limit of a random walk. In fact, this has been rigorously proven as Donsker's Invariance Principle. For DLA, this implies that large clusters formed by random walks should exhibit the same macroscopic properties as those formed by Brownian particles.*

The connection between the physical process (Brownian motion) and the geometric potential (Laplace's equation) is provided by the infinitesimal generator.

Definition 2.17 (Continuous Generator and Laplacian)**.** *The Laplace operator (or Laplacian) Δ in $\mathbb{R}^d$ is the sum of the second partial derivatives:*

$$\Delta = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}. \tag{2.34}$$

The infinitesimal generator of standard Brownian motion is the operator $\mathcal{A} = \frac{1}{2}\Delta$. It characterises the process via the martingale property: for any compactly supported C^2 (twice continuously differentiable) function f, the quantity

$$M_t = f(B_t) - f(B_0) - \int_0^t \frac{1}{2}\Delta f(B_s)\, ds \tag{2.35}$$

is a martingale with respect to the filtration of B_t.

This definition can be understood as a stochastic analogue to the Fundamental Theorem of Calculus. The integral term represents the expected drift of the function along the path. By subtracting this drift, we isolate the centred random noise, which is a martingale.

Remark 2.4 (Scaling Factors and Independence)**.** *We point out here that the scaling factor here differs from that in section 2.2.1. This discrepancy stems from the second-order term of the Taylor expansion used to derive the Brownian limit. However, since the potential distribution for DLA is determined by the condition $\mathcal{L}u = 0$ (or $\Delta u = 0$), this constant factor cancels out. The resulting harmonic measure is invariant. For a full derivation of this limit, see Lawler and Limic [5, Section 2.1].*

Definition 2.18 (Harmonic Functions)**.** *A twice continuously differentiable function $u : D \to \mathbb{R}$ defined on a domain $D \subset \mathbb{R}^d$ is called harmonic if it lies in the kernel of the Laplace operator:*

$$\Delta u(x) = 0 \quad \text{for all } x \in D. \tag{2.36}$$

By comparing this with the generator definition above, we see immediately that if u is harmonic, the drift term vanishes. This yields the critical result that $u(B_t)$ is a martingale, implying that the "average value" of a harmonic function is conserved along Brownian paths. We formalise this as the mean value property.

Theorem 2.6 (Mean Value Property)**.** *A locally integrable function $u : D \to \mathbb{R}$ is harmonic ($\Delta u = 0$) if and only if for every ball $B(x, r)$ strictly contained in the domain D:*

$$u(x) = \frac{1}{|\partial B(x,r)|} \int_{\partial B(x,r)} u(y)\, d\sigma(y), \tag{2.37}$$

where σ denotes the surface measure on the boundary sphere.

This is the continuous equivalent to the 'average of its neighbours' property of in the discrete setting.

2.3.2 Harmonic Measure and the Poisson Kernel

Before we present the continuous analogue to theorem 2.4, we define the continuous version of the harmonic measure.

Definition 2.19 (Harmonic Measure)**.** *Let $D \subset \mathbb{R}^d$ be a domain. The harmonic measure $\omega_x = \omega_x(D, \cdot)$ is the probability measure on the boundary ∂D defined by:*

$$\omega_x(E) = \mathbb{P}^x(B_{\tau_D} \in E), \tag{2.38}$$

for Borel sets $E \subset \partial D$, where $\tau_D = \inf\{t > 0 : B_t \notin D\}$ is the first exit time.

The solution to the continuous Dirichlet problem is given by Kakutani's Theorem.

Theorem 2.7 (Kakutani's Theorem)**.** *If D is a bounded domain and g is a continuous function on ∂D, then the function defined by:*

$$u(x) = \mathbb{E}^x[g(B_{\tau_D})] = \int_{\partial D} g(y)\, d\omega_x(y) \tag{2.39}$$

is harmonic in D and satisfies $\lim_{x \to y} u(x) = g(y)$ for regular boundary points $y \in \partial D$. Here, ω_x denotes the harmonic measure relative to x, representing the distribution of the first hitting point of Brownian motion on ∂D.

Proof. For a rigorous proof relying on the martingale property of harmonic functions, see Mörters and Peres [14, Theorem 3.12]. □

As in the discrete case, we are concerned with extending this to infinity. As a consequence of the following lemma, 2D Brownian motion is neighbourhood recurrent, meaning it will revisit any neighbourhood infinitely often. This prevents us from simply imposing a boundary condition at infinity, just as recurrence did in the discrete case.

Lemma 2.1 (Exit Probabilities from an Annulus)**.** *Let B_t be a complex Brownian motion starting at z_0 within the annulus $A = \{z \in \mathbb{C} : r < |z| < R\}$. Let $T_A = \inf\{t > 0 : B_t \notin A\}$ be the first exit time. The probability that the process exits through the inner boundary $|z| = r$ is given by:*

$$\mathbb{P}^{z_0}(|B_{T_A}| = r) = \frac{\ln R - \ln |z_0|}{\ln R - \ln r}. \tag{2.40}$$

Proof. See Cass [15, Lemma 43]. The proof uses Itô's formula (stochastic chain rule) on the function $\ln |Z_t|$. □

Corollary 2.1 (Recurrence)**.** *Planar Brownian motion is neighbourhood recurrent. For any starting point z_0 and any $\epsilon > 0$, the process will enter the disk $\{z : |z - z_0| < \epsilon\}$ almost surely:*

$$\mathbb{P}(\exists t > 0 : |B_t - z_0| < \epsilon) = 1. \tag{2.41}$$

Proof. Taking the limit $R \to \infty$ in the annulus exit probability yields $\mathbb{P}(|B_{T_A}| = r) \to 1$. □

2.3.3 Conformal Invariance

While in the discrete case we resolved this recurrence issue by turning to (regularised) Green's functions, we will instead utilise a fundamental property of Brownian motion: conformal invariance.

Theorem 2.8 (Lévy's Conformal Invariance Theorem)**.** *Let $U \subset \mathbb{C}$ be a domain and $f : U \to V$ be a non-constant analytic function. If Z_t is a complex Brownian motion in U, then the image process $W_t = f(Z_t)$ is, up to a random time change, a complex Brownian motion in V. Specifically, if $\tau_t = \int_0^t |f'(Z_s)|^2 ds$, then $f(Z_{\tau^{-1}(t)})$ is a standard Brownian motion in V.*

Proof. See Cass [15, Theorem 40]. □

Corollary 2.2 (Conformal Invariance of Harmonic Measure)**.** *Let $f : D \to D'$ be a conformal map. The harmonic measure is invariant under f:*

$$\omega_z(D, E) = \omega_{f(z)}(D', f(E)) \tag{2.42}$$

for any Borel set $E \subset \partial D$.

This result guarantees that utilising conformal maps in the sampling process will not distort the underlying harmonic measure, preserving the integrity of our off-lattice algorithm in the subsequent section.

2.3.4 Solvable Geometries

We conclude by introducing the probability density function for the hitting probabilities, and finding it explicitly for two fundamental domains used in DLA algorithms: the Upper Half-Plane ($\mathbb{H}$) and the Unit Disk ($\mathbb{D}$).

Definition 2.20 (Poisson Kernel)**.** *Let σ be the surface measure on ∂D. If the harmonic measure is absolutely continuous with respect to ζ, the Poisson kernel $P(x, y)$ is defined as the Radon-Nikodym derivative:*

$$P(x, y) = \frac{d\omega_x}{d\sigma}(y). \tag{2.43}$$

It represents the probability density of the Brownian motion exiting D at y given it started at x.

We can now express the conformal invariance of the harmonic measure in terms of the Poisson kernel. By definition, the harmonic measure of a set $E \subset \partial D$ is the integral of the Poisson kernel with respect to the surface measure $d\sigma$:

$$\omega_z(D, E) = \int_E P_D(z, \zeta) d\sigma(\zeta). \tag{2.44}$$

Applying the conformal invariance corollary to a map $f : D \to \mathbb{D}$ such that $f(z) = 0$, and noting that the Poisson kernel on the disk from the origin is uniform ($P_{\mathbb{D}}(0, w) = \frac{1}{2\pi}$), we have:

$$\int_E P_D(z, \zeta) d\sigma(\zeta) = \omega_0(\mathbb{D}, f(E)) = \int_{f(E)} \frac{1}{2\pi} d\sigma(w). \tag{2.45}$$

By the change of variables $w = f(\zeta)$, the arc length element transforms as $d\sigma(w) = |f'(\zeta)|d\sigma(\zeta)$, resulting in:

$$\int_E P_D(z,\zeta)d\sigma(\zeta) = \int_E \frac{1}{2\pi}|f'(\zeta)|d\sigma(\zeta). \tag{2.46}$$

This equality holds for any Borel set E, hence:

$$P_D(z,\zeta) = \frac{1}{2\pi}|f'(\zeta)|. \tag{2.47}$$

We now apply this relation to our domains of interest:

Theorem 2.9 (Poisson Kernel for $\mathbb{H}$)**.** *Consider the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : Im(z) > 0\}$. The probability density of a Brownian particle starting at $z = x + iy$ hitting the real line at $u \in \mathbb{R}$ is given by the Cauchy distribution:*

$$P_{\mathbb{H}}(z,u) = \frac{1}{\pi}\frac{y}{(u-x)^2 + y^2}. \tag{2.48}$$

Proof. We construct a conformal map $\phi : \mathbb{H} \to \mathbb{D}$ that sends the starting point z to the origin 0, and the real line $\mathbb{R}$ to the unit disk. The standard choice is the Cayley transform variation:

$$\phi(w) = \frac{w-z}{w-\bar{z}}. \tag{2.49}$$

Observe that $\phi(z) = 0$, and for any real boundary point $u \in \mathbb{R}$, $|\phi(u)| = 1$. Using the pullback proposition, the kernel is given by $\frac{1}{2\pi}|\phi'(w)|$. We compute the derivative with respect to the w:

$$\phi'(w) = \frac{z-\bar{z}}{(w-\bar{z})^2}. \tag{2.50}$$

Now evaluated at a boundary point $w = u$:

$$|\phi'(u)| = \frac{|2iy|}{|u-\bar{z}|^2} = \frac{2y}{(u-x)^2 + y^2}. \tag{2.51}$$

Thus, the Poisson kernel is:

$$P_{\mathbb{H}}(z,u) = \frac{1}{2\pi}\left(\frac{2y}{(u-x)^2 + y^2}\right) = \frac{1}{\pi}\frac{y}{(u-x)^2 + y^2}. \tag{2.52}$$

□

We now follow the same procedure for the unit disk.

Theorem 2.10 (Poisson Kernel for $\mathbb{D}$)**.** *The Poisson kernel for the unit disk $\mathbb{D}$, representing the exit density at angle θ for a particle starting at $z \in \mathbb{D}$, is given by:*

$$P_{\mathbb{D}}(z,\theta) = \frac{1}{2\pi}\frac{1-|z|^2}{|e^{i\theta} - z|^2}. \tag{2.53}$$

Proof. We seek an automorphism of the disk $\psi : \mathbb{D} \to \mathbb{D}$ that maps the starting point z to the

origin 0. This is given by the Möbius transformation:

$$\psi(w) = \frac{w - z}{1 - \bar{z}w}. \tag{2.54}$$

The derivative at the boundary point w is:

$$\psi'(w) = \frac{1 - |z|^2}{(1 - \bar{z}w)^2}. \tag{2.55}$$

To find the magnitude $|\psi'(w)|$, we observe that for $w = e^{i\theta}$, the denominator satisfies:

$$|1 - \bar{z}e^{i\theta}| = |e^{-i\theta} - \bar{z}| = |\overline{e^{-i\theta} - \bar{z}}| = |e^{i\theta} - z|. \tag{2.56}$$

Therefore, the Poisson kernel is:

$$\frac{1}{2\pi}|\psi'(e^{i\theta})| = \frac{1}{2\pi}\frac{1 - |z|^2}{|e^{i\theta} - z|^2}. \tag{2.57}$$

□

Corollary 2.3 (Poisson Kernel for the Exterior of $\mathbb{D}$)**.** *Let $D_{ext} = \{z \in \mathbb{C} : |z| > 1\}$. The harmonic measure of the unit circle relative to a point $z \in D_{ext}$ is given by the density:*

$$P_{ext}(z, \theta) = \frac{1}{2\pi}\frac{|z|^2 - 1}{|z - e^{i\theta}|^2}. \tag{2.58}$$

Proof. Consider the geometric inversion $a = 1/\bar{z}$, which maps $D_{ext} \to \mathbb{D}$. By the conformal invariance of harmonic measure, the exit distribution for z is identical to that of a. Substituting a into the interior kernel (theorem 2.10) and simplifying yields the result. □

This concludes the continuous formulation of the DLA process. The theoretical framework parallels the discrete setting but utilises slightly different machinery:

1. The system is governed by a continuous potential field $u(x)$ determined by the geometry of the cluster, representing the probability density of the underlying Brownian motion.

2. The field $u(x)$ satisfies the continuous Laplace equation $\Delta u = 0$, but solving this explicitly is difficult. Instead we utilise the property of conformal invariance, allowing us to encode the potential field geometrically.

3. The growth probability at a boundary point is given by the Poisson Kernel, which serves as the continuous analogue to the discrete flux.

Chapter 3

Algorithms

3.1 Introduction

Simulating DLA presents a significant computational challenge. A naive implementation, where a walker moves one lattice step at a time from a large distance, is prohibitively expensive; the number of steps required for a walker to hit a cluster of radius R scales as $O(R^2)$. To generate large clusters (e.g., $N = 10^7$ particles) required for useful analysis, we must employ accelerated algorithms that exploit the potential theory derived in chapter 2.

In this chapter, we develop three distinct implementation models:

1. **On-Lattice DLA:** A high-precision discrete implementation adapted from bias-free algorithm by Loh [16].

2. **Off-Lattice DLA:** A continuous implementation utilising conformal mapping techniques adapted from Bell [17].

3. **Hybrid DLA:** A novel semi-discrete model designed to decouple transport dynamics from aggregation geometry.

Despite the differing domains ($\mathbb{Z}^2$ vs $\mathbb{R}^2$), all three algorithms share a common three-stage architecture: Initialisation, Diffusion, and Aggregation. We analyse each stage in turn.

3.2 Initialisation

The mathematical definition of DLA requires particles to originate "from infinity." In practice, we must approximate this by launching particles from a boundary $\partial\Omega$ sufficiently far from the cluster K.

3.2.1 Off-Lattice: The Launching Circle

In the continuous setting, initialising at infinity is mathematically equivalent to initialising uniformly on a large bounding circle. Let R_{max} be the radius of the smallest circle centred at the origin containing the cluster. We define a launching circle of radius $R_l > R_{max}$.

Consider the conformal map $\phi(z) = \frac{R_l}{z}$. This creates a bijection between the exterior of the launching circle and its interior, mapping infinity to the origin. By the Mean Value Property, the

harmonic measure of the origin in the disk is uniform (theorem 2.10 with $z = 0$). Consequently, the hitting distribution of a Brownian motion starting at infinity on the circle $|z| = R_{launch}$ is uniform. Thus, our off-lattice algorithm simply samples a starting angle $\theta \sim U[0, 2\pi)$ and initialises the walker at $z_0 = R_l e^{i\theta}$. This is therefore exactly equivalent analytically to starting "at infinity". Note also that provided the circle contains the entire cluster, there is no size restriction, so we scale it dynamically.

3.2.2 On-Lattice: The Fuzzy Annulus

In the discrete setting, the "launching circle" approach introduces a subtle but significant bias. As detailed by Loh [16], analytically projecting a continuous circle onto a discrete grid forces the starting positions to "snap" to the nearest integer coordinates. This discreteness creates a 'lattice aliasing' effect (also referred to as a Moireé interference pattern), resulting in a bias. For clusters of size $N > 10^5$, this bias is sufficient to distort the fractal dimension.

To eliminate this, we implement Loh's Fuzzy Annulus method. Instead of a fixed radius R, the starting position is sampled from a probability distribution spread over an annulus. This probability distribution utilises the Kaiser-Bessel function I_0, which was developed to suppress high-frequency aliasing artifacts in signal processing [18]:

$$P(x) \propto \frac{I_0(x)}{I_0(R_l)}. \tag{3.1}$$

By sampling from this "fuzzy" distribution, the grid aliasing effects interfere destructively and cancel out, reducing the initialisation bias to negligible levels.

Like in the off-lattice case, we can scale the launching annulus dynamically, but to further reduce the bias, the inner boundary of the annulus is at $2R_{max}$.

Remark 3.1 (Rejection Sampling)**.** *Rather than directly sampling from this distribution, we use an optimisation technique called 'rejection sampling', which we explain in the appendix.*

3.3 Far-Field Diffusion

Once initialised, the walker is far from the cluster. Simulating every step of a random walk is inefficient. We therefore use skipping strategies based on the properties of harmonic functions.

3.3.1 Off-Lattice: Re-injection from Infinity

If the walker strays outside the launching circle ($|z_{out}| > R_l$), we exploit the conformal invariance established in section 2.3.

Rather than discarding the particle or simulating the return trajectory step-by-step, we sample the first hitting point on the circle directly. As shown in corollary 2.3, the return probability is governed by the exterior Poisson kernel. To sample from this efficiently, we avoid numerical integration of this density function. Instead, we directly utilise the conformal invariance of Brownian motion. This principle establishes a statistical equivalence between geometries: to sample the hitting distribution on a physical boundary $\partial\Omega$, we can sample the hitting distribution on a reference domain ∂D (where the distribution is known, e.g., Uniform) and transform the

result via the conformal map $f : D \to \Omega$. In this instance, we map the uniform distribution of the unit circle to the exterior harmonic measure via the following steps:

1. Normalisation: We normalise the walker's position to $r = |z_{out}|/R_l$ and map it to the interior point $a = 1/r$ via inversion.

2. Möbius Transformation: We generate a uniform variate ζ on the unit circle (representing an isotropic exit from the origin). We then apply the real-axis Möbius map:

$$w = \frac{r\zeta + 1}{r + \zeta}. \tag{3.2}$$

3. Reconstruction: The resulting point w is distributed exactly according to the exterior harmonic measure. We rotate w by $\arg(z_{out})$ and scale by R_l to obtain the true re-entry position.

This effectively folds the infinite exterior space back onto the launch circle in $\mathcal{O}(1)$ operations.

Remark 3.2. *This 'no-killing' approach was formally introduced by Menshutin and Schur [19] as a 'killing-free' algorithm, where they sampled from the exact hitting distribution (via the Poisson Kernel). We instead adopt Bell's conformal mapping approach, although we map to the unit disk while Bell maps to a straight line, which should make our code very slightly more computationally efficient.*

3.3.2 On-Lattice: Walk-to-Line

In the discrete setting, we cannot simply use a conformal map to transport a walker from infinity to the cluster's boundary, as the grid lacks the continuous symmetries required for conformal invariance. As an alternative, we employ Loh's Walk-to-Line acceleration method.

When a walker wanders outside the cluster's bounding box, we project its position onto the nearest face of the box. We treat this face as an infinite absorbing line $\partial\Lambda = \{(x, 0) : x \in \mathbb{Z}\}$ and the walker's position as $z_0 = (0, h)$ in the upper half-plane $y > 0$. The problem reduces to sampling the exit probability on the line, the discrete analogue to theorem 2.9.

3.3.2.1 Discrete Method of Images

As established in chapter 2, the exit probability (harmonic measure) at a boundary point $(x, 0)$ is proportional to the discrete flux into that point. Since the potential $u(z)$ vanishes on the boundary, the flux is simply the value of the potential at the immediate interior neighbor:

$$P_{\text{lattice}}(x) \propto \Phi(x, 0) = u(x, 1). \tag{3.3}$$

To calculate this, we must construct a potential $u(z)$ that is harmonic in the upper half-plane but strictly satisfies the Dirichlet boundary condition $u(x, 0) = 0$ (representing the absorbing line). We achieve this via the Method of Images.

The fundamental idea is to place a fictitious "negative walker" (or sink) at the reflected position $(0, -h)$, opposite the true walker at $(0, h)$. Due to the symmetry of the lattice, the potential generated by the real walker at any point on the boundary line $y = 0$ is identical to the

potential generated by the image. By subtracting the image potential from the source potential, these contributions cancel out exactly along the line, ensuring $u(x, 0) = 0$ by construction.

Using the discrete potential kernel $a(z)$ defined in section 2.2.3, the resulting field is:

$$u(x, y) = a(x, y + h) - a(x, y - h). \tag{3.4}$$

Substituting this into the flux equation ($y = 1$), we obtain the exact unnormalised probability of the walker hitting the line at relative coordinate x:

$$P_{\text{lattice}}(x) = a(x, 1 + h) - a(x, 1 - h). \tag{3.5}$$

Remark 3.3. *To implement this, we follow Loh exactly. For $h \leq 60$, we evaluate this using his pre-calculated lookup table of $a(z)$. For larger distances, we use the asymptotic expansion of the potential kernel, given in the appendix of [16].*

Remark 3.4. *Computing the normalisation constant for $P_{lattice}(x)$ over the infinite line is infeasible. Instead, we once again employ rejection sampling (details in appendix A.3).*

3.3.2.2 The Killing Circle

While the Walk-to-Line method is theoretically valid for any distance, practical implementations are bound by the limits of 64-bit integer arithmetic. If a walker wanders exceptionally far from the cluster (e.g., $R > 10^{14}$ lattice units), the coordinate calculations may overflow.

To prevent numerical instability, we define a hard "Killing Radius" $R_{kill} = 10^{14}$. In the extremely rare event that a walker's random step projects it beyond this limit, the particle is discarded and a new one is initialised. Given the recurrence of 2D random walks, this truncation introduces a vanishingly small statistical error ($\ll 10^{-10}$), which is acceptable for ensuring numerical robustness.

3.4 Near-Field Diffusion

As the walker approaches the cluster, the step size must decrease to avoid overshooting the boundary.

3.4.1 Off-Lattice: Walk-to-Circle

At each step, we calculate the shortest Euclidean distance to the cluster, d_{min}. We can construct a circle C of radius $r = d_{min} - \epsilon$ (where ϵ is a small tolerance). Since the interior of C is empty, the potential is harmonic. By the isotropy of Brownian motion (theorem 2.10 with $z = 0$), the exit distribution is uniform. We therefore advance the walker:

$$z_{new} = z + re^{i\theta}, \quad \theta \sim U[0, 2\pi). \tag{3.6}$$

This allows the walker to adaptively skip over large empty regions, taking large steps in the void and small steps in the fjords.

3.4.2 On-Lattice: Walk-to-Square

In the discrete setting, circles do not map cleanly to the grid. Instead, we employ Loh's Walk-to-Square method. This relies on the pre-calculation of exit probabilities for a hierarchy of empty square sizes $L_k = 2^k$ (e.g., from 4×4 up to 512×512).

Unlike a circle, the exit distribution on the boundary of a square is not uniform, as lattice points near the mid-edges are hit more frequently than the corners. We determine these exact probabilities by solving the discrete Dirichlet problem $\mathcal{L}u = 0$ for each square size S_k. For simplicity we used Loh's precomputed solutions (after verification).

Remark 3.5 (Alias Sampling)**.** *To sample from these non-uniform distributions efficiently, we calculate them offline and store them using Walker's Alias Method (detailed in appendix A.4). This allows us to sample the discrete exit location in $\mathcal{O}(1)$ time, regardless of the square size.*

The simulation step proceeds as follows:

1. **Scan:** Determine the largest power-of-two square S_k centred at the walker that does not intersect the cluster.
2. **Sample:** Query the pre-computed Alias Table for size k to generate a boundary coordinate z_{exit}.
3. **Jump:** Update the walker's position to z_{exit} instantly.

This allows the walker to leap across large voids while preserving the exact non-uniform statistics of the lattice random walk.

3.5 Aggregation

The final stage is the termination of the walk.

3.5.1 On-Lattice: Sticky Boundary

The aggregation condition on $\mathbb{Z}^2$ is straightforward. We define the boundary ∂K as the set of all empty sites adjacent to occupied sites, as we did in chapter chapter 2. The walk terminates when the particle steps onto a site $x \in \partial K$. The site x is then marked as occupied ($x \in K$), and the boundary ∂K is updated locally.

3.5.2 Off-Lattice: Exact Adhering

In the continuous model, aggregation occurs when the walker touches a cluster particle (i.e., the center-to-center distance is 2, since the radius of each particle is 1). Simulating this directly with fixed time-steps is not ideal, due to the potential errors of the walker grazing the cluster or overshooting.

To resolve this, we adopt the analytic method described by Bell [17]. When the walker is at position p, close to a target particle centered at q but constrained by a second-nearest neighbour q_2 at distance a, we define a local domain Ω. As illustrated in fig. 3.1 (Stage 1), Ω is the lune

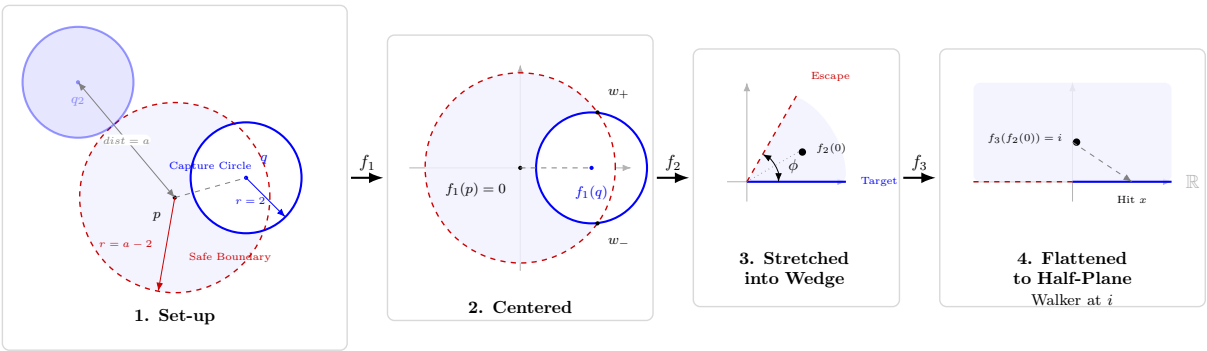


Figure 3.1: The conformal mapping sequence for exact collision. The physical domain (1), formed by the intersection of the safe region and the target exterior, is centred (2), mapped to an infinite wedge (3), and finally opened to the upper half-plane (4) where the exit position is sampled.

formed by the intersection of the "safe escape circle" (radius $a-2$, centered at p) and the exterior of the "capture circle" (radius 2, centered at q).

To sample the exact exit position from Ω without time-stepping errors, we conformally map this lune to the upper half-plane $\mathbb{H}$ in three stages:

1. **Centering (f_1):** We first translate and rotate the system so that the walker p lies at the origin and the target q lies on the positive real axis:

$$f_1(z) = |q-p|\frac{z-p}{q-p}. \tag{3.7}$$

   As shown in fig. 3.1 (Stage 2), we identify the intersection points of the two circles in this transformed space, denoted w_+ and w_-.

2. **Stretching (f_2):** We apply a Möbius transformation that maps the intersection point w_+ to the origin and w_- to infinity:

$$f_2(z) = \frac{z - f_1(w_+)}{z - f_1(w_-)}. \tag{3.8}$$

   This transformation stretches the circular boundaries into two straight rays emanating from the origin, forming an infinite wedge (fig. 3.1, Stage 3). The angle ϕ between these rays is determined by the intersection geometry. The walker now lies somewhere on the segment of the unit circle within this wedge.

3. **Flattening (f_3):** Finally, we open the wedge into the full upper half-plane using a power map:

$$f_3(z) = z^{\pi/\phi}. \tag{3.9}$$

   This maps the target boundary to the positive real line and the escape boundary to the negative real line (fig. 3.1, Stage 4). Due to the symmetry of the original lune about the $p-q$ axis, the walker is mapped to the point i.

In this final domain, the problem reduces to a standard Brownian motion starting at i and hitting the real line. We sample the hitting point from the Cauchy distribution (derived in section 2.3.4, which easily integrates to give the CDF) and map it back to the physical domain

using the inverse composition $f_1^{-1} \circ f_2^{-1} \circ f_3^{-1}$. If the mapped point lies on the target circle, the particle adheres; if it lies on the safe circle, the walker is effectively reset, and diffusion continues.

3.6 The Hybrid Model

To rigorously disentangle the source of anisotropy in DLA, we introduce a hybrid model.

3.6.1 Motivation and Design

We hypothesise that the characteristic cross-shape of on-lattice DLA arises primarily from the restriction of particles to fixed grid sites rather than the transport dynamics. To test this, we develop a model where the transport is purely isotropic (Brownian motion), but the final aggregation is constrained to a grid geometry.

The critical design choice is the aggregation mechanism. We evaluated three potential strategies:

1. The Sticky Boundary: A walker aggregates if it enters any square cell within the sticky boundary. We rejected this because the square capture zone imposes lattice anisotropy onto the incoming probability flux before aggregation occurs, preventing a clean separation of transport and packing effects.

2. The Antennae Method: As proposed by Menshutin and Shchur [10], particles only stick if they hit a lattice site along specific axes; otherwise, they are reflected. This is similar in some sense to "Noise-Reduced DLA" [20] which was designed to accelerate convergence to asymptotic forms. We don't use this method as it effectively creates an anisotropic sticking probability ($P_{stick} < 1$ for off-axis collisions), which could potentially prevent us from isolating the effects of the packing geometry.

3. Rotate and Snap **(Chosen)**: The walker behaves exactly as a hard sphere in continuous space. Upon collision, the particle relaxes into the nearest valid lattice site. This method preserves the exact continuous harmonic measure of the cluster until the instant of impact, ensuring the arrival statistics are purely isotropic.

Remark 3.6. *One might question whether the specific choice of aggregation rule significantly alters the growth. Theoretical estimates suggest that these microscopic differences are likely negligible.*

As derived from Beurling's estimates on harmonic measure [21], once a random walker enters the immediate vicinity of a boundary point, the probability of it escaping to hit a different, distant site decays rapidly. Specifically, if a walker is at a small distance δ from a cluster site P, the harmonic probability ω that it diffuses away to hit a different site Q (at distance $r \gg \delta$) is bounded by:

$$\omega(z \to Q) \lesssim \left(\frac{\delta}{r}\right)^{\alpha} \tag{3.10}$$

where $\alpha > 0$ depends on the local geometry.

This implies that the decision of where to aggregate is effectively determined by the macroscopic flux before the walker resolves the microscopic shape of the target site. Therefore the exact adherence

rule is likely not overly important, although we still prefer our method for the reasons outlined previously.

3.6.2 Implementation

The algorithm proceeds identically to the Off-Lattice simulation (Walk-on-Spheres) until the collision condition $|z_{walker} - z_{cluster}| \leq 2$ is met (since particle radius $r_p = 1$).

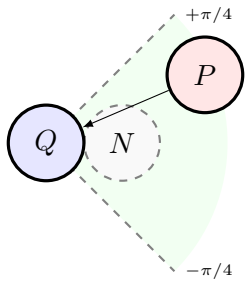


Figure 3.2: The angular window for hybrid aggregation. If a walker P collides with particle Q within the angle $|\phi| < \pi/4$ (shaded region), it is snapped to the nearest lattice slot N. We can assume N is empty because if it were occupied, its own collision shell would have intercepted the walker before it could reach Q within this specific angle.

Let the continuous collision angle relative to the target particle center be $\phi \in [0, 2\pi)$. We project this continuous angle onto the discrete set of lattice directions $\Theta = \{0, \pi/2, \pi, 3\pi/2\}$ by minimizing the angular distance:

$$\theta_{snap} = \frac{\pi}{2} \left\lfloor \frac{4\phi + \pi}{2\pi} \right\rfloor . \tag{3.11}$$

The particle is then rigidly placed at the lattice coordinate:

$$z_{new} = z_{cluster} + 2e^{i\theta_{snap}}. \tag{3.12}$$

A potential concern is whether the "snapped" lattice site is available. We provide a simple geometric argument to justify that the target slot is always empty.

Consider a walker P hitting a cluster particle Q at angle ϕ. The snapped destination is the lattice neighbour N closest to the point of impact. For a grid of spacing 2, the neighbour N is located at distance 2 from Q (centre-to-centre). If N were already occupied, its own collision shell (radius 2) would overlap the collision shell of P.

Simple trigonometry shows that for any impact angle ϕ closer to N than any other neighbour (i.e., $|\phi| < \pi/4$), the distance from the impact point to N is strictly less than 2:

$$|z_{hit} - z_N| = |2e^{i\phi} - 2| = 2\sqrt{2 - 2\cos\phi}. \tag{3.13}$$

For $|\phi| < \pi/4$, this distance is at most $2\sqrt{2 - \sqrt{2}} \approx 1.53$, which is well inside the collision radius of 2. This implies that if N were occupied, the walker would have intersected N's collision shell before reaching P's surface. Since the algorithm halts at the first contact, the target slot N must be empty.

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Appendix A

Implementation Details and Optimisations

While chapter 3 outlines the mathematical principles of the algorithms, the practical implementation requires specific data structures and optimisations to achieve the necessary scale. This appendix details the computational architecture of the simulation code, written in Python and accelerated using the Numba JIT (Just-In-Time) compiler.

The lattice and off-lattice versions of the algorithm are versions of code created by Loh [16] and Bell [17] respectively. However, these were originally done in C and C++, and in translating to Python our implementation diverges both to account for differences in the language structure, and to account for the limitations of our computer (a 2017 Macbook with 8GB RAM). We will highlight some of these changes here, as well as explain some of the more programming-related concepts and structures behind the algorithms.

The full source code for all models and analysis scripts is available at `https://github.com/RamsayDavis/dla`.

A.1 Computational Framework

A.1.1 The Numba JIT Compiler

The simulations are implemented in Python to leverage its flexibility, to consistent with the analysis section of this project, and as it is the language we are most comfortable with. The core numerical routines are decorated with `@njit(fastmath=True)`. This compiles the Python bytecode into optimised LLVM machine code at runtime, allowing for performance parity with C++ while maintaining a modular, functional codebase. Our code is likely somewhat slower than the original implementations, but this mitigates this difference, and allows us to generate clusters of size 10^7 particles in about 10 to 15 minutes on our CPU.

A.1.2 Rounding Semantics

One minor but important detail to consider in stochasti simulations is the rounding behaviour of floating-point co-ordinates, particularly in the lattice and hybrid models where the output is in integers.

- Python: Python 3 employs "Banker's Rounding" (round half to even). For example, `round(2.5) -> 2` and `round(3.5) -> 4`. This introduces a subtle statistical bias in random walks, as even coordinates are slightly favoured over odd ones.
- C and C++: The reference C++ implementations use the standard library `round()` function, which rounds half away from zero (e.g., `round(2.5) -> 3`).

To ensure our random walks match the theoretical lattice statistics and the reference datasets exactly, we implemented a custom kernel `_round_cpp(x)` that explicitly emulates the C++ behaviour in Python.

A.2 Spatial Indexing and Hierarchical Skipping

In the diffusion steps of all versions of DLA, we require the distance from the walker to the nearest particle in the cluster, in order to work out the maximum allowed step-size. Additionally it is needed in the off-lattice algorithm for the adhesion step, since we need the two nearest particles.

To do this efficiently, we employ a Hierarchical Map. The exact implementation differs between the two models, but the essential idea is we overlay different sized grids over the cluster and surrounding area, and record whether the cells in each grid are occupied or not. By traversing the different grid layers, we can quickly work out these distances.

A.2.1 Lattice: Flattened Bit-Map

Data Structure: In the discrete model, the hierarchy must have enough layers to view individual lattice sites at the finest level. Loh's code uses a 2D array of pointers, one per hierarchy level. However for large grids this can be problematic in Python, with memory fragmentation and cache locality issues.

We instead implement a Flattened 1D Bit-Map. All levels are packed into a single contiguous `uint8` array, `flat_ss`, indexed via an offset table. We track occupancy at the bit level (1 bit per site).

Linear Search: We utilise pre-computed Green's function tables for jumps of radius 2^0 to 2^8 (corresponding to the squares of sizes 2 to 512). The routine `_diffuse_variable_flat` performs a linear scan starting from the coarsest useful level (pixel size ≈ 256) down to the finest:

1. Start: Check level $L_{start} = \max(2, L_{max} - 9)$.
2. Check: If the 3×3 neighbourhood is empty, the walker is in a void of radius ≥ 256. We immediately apply the maximum jump ($R = 256$) and exit.
3. Repeat: If occupied, we increment the level index to check finer resolutions ($R = 128, 64 \ldots 1$).

While L_{max} is often greater than 9, we only have precalculated tables for up to this size, hence the restriction. A potential improvement to the on-lattice code would be to include pre-calculated tables of larger sizes, but this was not a priority for achieving our goal cluster sizes.

A linear scan is preferred over binary search here because DLA walkers spend most time in voids, where the first (coarsest) check succeeds immediately.

A.2.2 Off-Lattice: Dual-Structure Strategy

The continuous model separates "Skipping" and "Hitting" into two distinct structures to optimise for dynamic memory usage.

A.2.2.1 1. The Coarse Hierarchy (Skipping)

Structure: A dynamic list of 2D arrays. Unlike the lattice version, we do not flatten this into a single array. At the finest level, the mesh width is approximately 24 units, meaning the memory pressure is orders of magnitude lower than the unit-resolution lattice grid.

Binary Search: Since continuous walkers can jump any distance, we utilise the full depth of the hierarchy. We use a binary search to efficiently locate the transition layer between "Occupied" and "Empty". This guarantees $\mathcal{O}(\log L)$ performance, ensuring stability whether the walker is in a void or a deep fjord. The greater number of layers makes a binary search worthwhile here.

A.2.3 2. The Spatial Hash (Hitting)

Structure: When the hierarchy indicates the walker is close to the cluster, we require the exact particle coordinates for the adhesion step. However, the coarse hierarchy only records occupancy and not the precise coordinates. To query nearest neighbours in $\mathcal{O}(1)$ time without dynamic memory allocation, we implement a static chained hash table on a fine grid.

This mimics a “List of Lists" using two pre-allocated integer arrays:

1. `grid_head[cell_index]`: Stores the index of the most recently added particle in the grid cell. Initialised to -1.
2. `particle_next[particle_index]`: Stores the index of the next particle in the same cell, forming a linked list.

Insertion: When a new particle P_{new} is added to cell C:

$$\texttt{particle_next}[P_{new}] = \texttt{grid_head}[C] \quad ; \quad \texttt{grid_head}[C] = P_{new} \tag{A.1}$$

Bell’s code used C++ vectors, with one vector per cell in the finest layer of the hierarchy. We prefer our linked list approach as we avoid resizing arrays or allocating objects. In both cases we can iterate through the (linked) list to find the closest particle to the walker.

A.3 Rejection Sampling

A.3.1 Concept and Motivation

Several probability distributions required for the on-lattice simulation possess Cumulative Distribution Functions (CDFs) that are computationally expensive or impossible to invert analytically.

A naive approach would be to discretise the domain and build a lookup table (numerical integration). However, this introduces systematic discretisation error. Instead, we employ rejection sampling. This method allows us to generate samples from a complex target distribution $f(x)$ using a simpler "proposal" distribution $g(x)$, producing exact samples from $f(x)$ without any approximation error.

A.3.2 The Algorithm

Given a target density $f(x)$ and an envelope density $g(x)$ such that $f(x) \leq Mg(x)$ for all x (where M is a constant scaling factor), the algorithm proceeds as follows:

Algorithm 1: Rejection Sampling

```
Function SampleRejection(f, g, M):
    while True do
        X ← Sample(g) ;                          // Propose from envelope
        U ← RandomFloat(0, 1) ;                  // Verification variable
        if U ≤ f(X) / (M·g(X)) then
            return X ;                           // Accept and return
        // Else: Reject and loop again
```

Geometrically, this is equivalent to generating uniform random points under the curve $Mg(x)$ and keeping only those that fall under the curve $f(x)$. While the runtime is non-deterministic (dependent on the acceptance rate $1/M$), it is often computationally superior to numerical inversion because it avoids evaluating expensive special functions (like integrals of Bessel functions) for every rejected sample.

A.3.3 Application 1: The Fuzzy Annulus

To eliminate lattice aliasing, we launch particles from a radial distribution proportional to the Kaiser-Bessel function I_0.

- $f(x)$**:** $I_0(\beta\sqrt{1-x^2})$.
- $g(x)$**:** A Gaussian distribution $\mathcal{N}(0, \sigma^2)$ that strictly bounds I_0.

We sample candidate radii from the Gaussian (using standard library functions) and reject them based on the ratio of the Bessel weight to the Gaussian weight.

A.3.4 Application 2: Walk-to-Line

When a walker projects onto the bounding box face, we must sample the exit position on the line $y = 0$.

- $f(x)$**:** The exact discrete flux calculated via the Method of Images ($P_{lattice} \propto u(x, 1) - u(x, -1)$).
- $g(x)$**:** The Cauchy distribution (the analytical solution for continuous Brownian motion).

As established by Loh [16], the continuous flux serves as a valid envelope for the discrete flux. By setting the scaling constant $M \approx 1$, we ensure the condition $f(x) \leq Mg(x)$ holds, resulting in a computationally efficient sampling routine with a high acceptance rate.

A.4 Walker's Alias Method

A.4.1 Motivation

For the "Walk-to-Square" algorithm, we must sample the exit location of a walker from the boundary of a square. This distribution is non-uniform (the exit probability is highest at the midpoints of the edges and lowest at the corners).

A naive approach be to search a Cumulative Distribution Function (CDF). For a square of perimeter N, this requires a binary search, scaling as $\mathcal{O}(\log N)$. As the square size increases (e.g., $N = 2048$ for a 512×512 square), this $\log N$ cost inside the inner loop becomes a significant bottleneck.

To eliminate this overhead, we follow Loh and implement Walker's Alias Method, which reduces the sampling complexity to $\mathcal{O}(1)$—constant time, regardless of the square size.

Intuitively, the Alias Method works by reshaping the uneven probability histogram into a perfectly rectangular block.

Imagine the probability mass function as a set of vertical bars. We normalize them so the average height is 1. The method then "slices" the excess height from "over-full" bins (where $P > 1$) and moves these blocks to fill the gaps in the "under-full" bins (where $P < 1$).

The result is a table of size N where every bucket contains exactly the same total probability mass ($1/N$), but that mass is shared between at most two outcomes: the original occupant of that bucket, and the overflow from one other bin.

A.4.2 Implementation

We pre-compute two lookup tables for every square size $L = 2^k$:

- **Prob Table (U_i):** The threshold probability for keeping the original index i.
- **Alias Table (K_i):** The alternate index if the threshold check fails.

Sampling is then performed by first a discrete uniform random sample, and then a continuous uniform sample which is compared to the threshold value:

Algorithm 2: Alias Sampling (Constant Time)

```
Function SampleAlias(U, K):
    i ← RandomInteger(0, N − 1) ;          // Select a bucket uniformly
    y ← RandomFloat(0, 1) ;                // Select height within bucket
    if y < U[i] then
        return i ;                         // Return bucket owner
    else
        return K[i] ;                      // Return bucket alias
```

A.5 Numerical Approximations

A.5.1 Custom Bessel Functions

The Kaiser-Bessel launching distribution requires the Modified Bessel Function of the first kind, $I_0(x)$. The standard Python implementation, `scipy.special.i0`, is not supported in Numba's `nopython` mode (as it links to C Fortran libraries not visible to the JIT compiler). We therefore implemented the polynomial approximation from Abramowitz and Stegun [22] (Eq. 9.8.1) directly in Python/Numba. This ensures that the launching bias correction is applied efficiently without breaking the JIT compilation chain or requiring callbacks to the Python interpreter.

A.6 Usability and Data Management

To facilitate large-scale parameter sweeps and ensure consistency between the discrete and continuous datasets, we introduced two workflow improvements to the reference implementation.

A.6.1 Heuristic Grid Sizing

The spatial extent of a DLA cluster is stochastic and difficult to predict a priori. Loh's code requires the user to manually define the grid dimension L_{max}. Underestimating this parameter leads to runtime failures (truncation), while overestimating it increases memory overhead.

To automate this, we implemented a heuristic initialisation step. Before memory allocation, we estimate the expected cluster radius R_{est} using the known fractal dimension $D_f \approx 1.71$ and apply a safety factor of 1.5 to account for the irregularity of the dendritic arms:

$$L_{max} = \lceil \log_2(1.5 \times 2.2 \times N^{1/1.71}) \rceil \tag{A.2}$$

This ensures the simulation domain dynamically scales to accommodate the target particle count N without manual intervention.

A.6.2 Standardised Coordinate Output

The internal state representations of the two models are fundamentally different: the lattice model uses a boolean bit-map of occupancy, while the off-lattice model uses complex floating-point numbers with a particle radius $r = 1$ (effective diameter $d = 2$).

To enable a unified analysis pipeline, we implemented a standardisation layer that normalises both outputs into an identical format: a centered $(N, 2)$ array of dimensionless coordinates.

- **Lattice Conversion:** We convert the sparse bit-map into a list of integer coordinates representing the centers of the occupied sites, shifting the seed to the origin $(0, 0)$. We also ensure these are only the occupied cells, as Loh's code includes the sticky sites as well.

- **Off-Lattice and Hybrid Normalisation:** We rescale the continuous coordinates by a factor of 0.5. Since the off-lattice physics uses $d = 2$ and the lattice uses unit spacing $d = 1$, this normalisation ensures that spatial metrics (such as the Radius of Gyration) are directly comparable across models.

This post-processing step allows us to apply the exact same analysis scripts (fractal dimension estimation, radial density functions) to all datasets agnostically, eliminating potential errors arising from manual unit conversion.

Appendix B

Reference: LaTeX Structure Examples

This section provides templates for common mathematical structures used in this dissertation.

B.1 Aligned Equations

Below is a multi-line derivation showing the alignment of the equals sign using the `align` environment.

$$\begin{aligned} \nabla^2\phi(r,\theta) &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} && \text{(B.1)} \\ &= 0 \quad \text{(Laplace Equation)} \\ \implies \phi(r,\theta) &= \sum_{n=0}^{\infty} r^n (A_n \cos n\theta + B_n \sin n\theta). && \text{(B.2)} \end{aligned}$$

B.2 Floats

B.2.1 Single Figure

Figure B.1 shows a standard inclusion of a graphic.

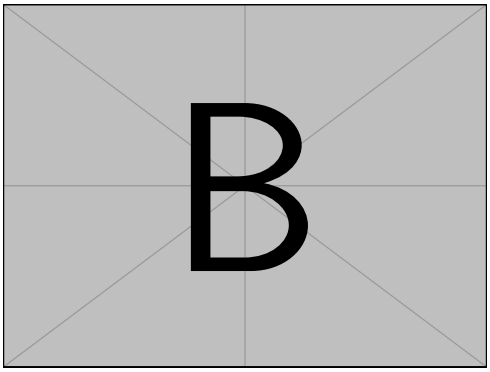
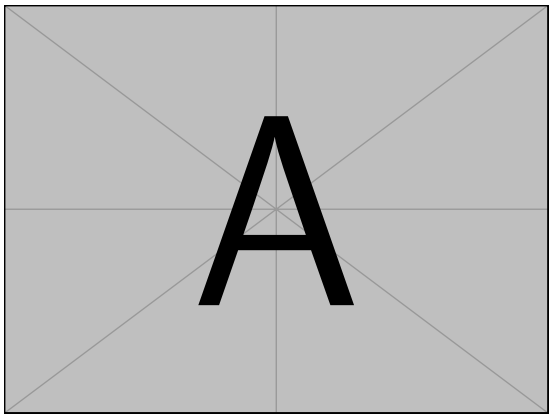


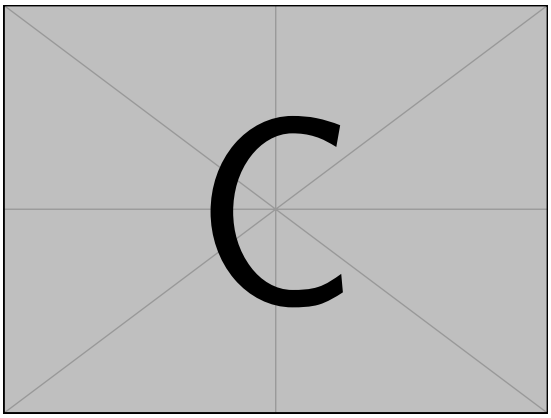
Figure B.1: A realization of 2D Brownian motion.

B.2.2 Side-by-Side Figures (Subfloats)

Figure B.2 demonstrates how to place two plots next to each other for comparison.



(a) On-Lattice DLA



(b) Off-Lattice DLA

Figure B.2: Comparison of cluster morphology between lattice and continuum models.

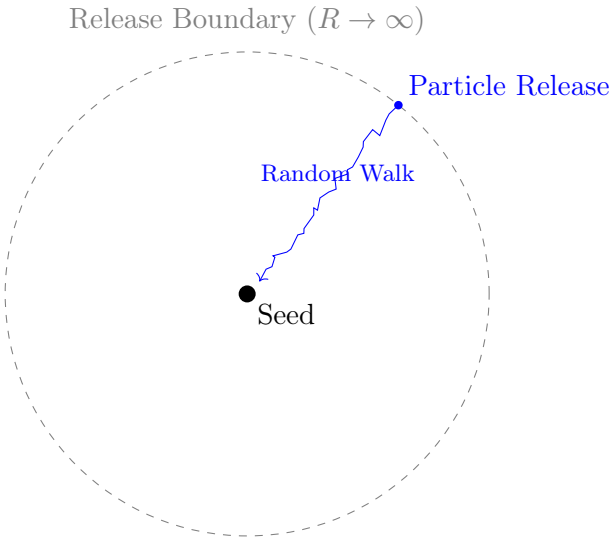


Figure B.3: Schematic of the DLA growth process.

B.3 Algorithms

Algorithm 3: Generalised DLA Simulation (Single Particle)

```
Data: Cluster C, Launch Radius R_launch, Killing Radius R_kill
Result: Updated Cluster C
Generate particle position r on launch boundary (|r| ≈ R_launch)
while particle is active do
    d ← distance from r to nearest point in C
    if distance to centre > R_kill then                    // Far Field
        Respawn particle
    else if d is outside bounding box then                 // Outer Vacuum
        r ← Apply Return-to-Boundary
    else if d > local step limit then                      // Empty Space
        r_jump ← max(safe step, d − δ)
        Move r by r_jump in random direction
    else                                                   // Interaction Zone
        Perform local adhesion step (lattice, continuous or hybrid)
        if Aggregation Condition Met then
            Add particle to C
            Update Cluster Radius R_max
            break
        end
    end
end
```

Algorithm 4 shows a basic random walk loop using the `algorithm2e` package.

Algorithm 4: Basic Random Walk Simulation

```
Result: Final position of particle
Initialize particle at r_start;
while particle not stuck AND particle within bounds do
    Generate random angle θ ∈ [0, 2π);
    x ← x + δ cos θ;
    y ← y + δ sin θ;
    if distance to cluster < ϵ then
        Stick particle;
        break;
    end
end
```

B.4 Theorems

Standard environments for formal mathematical statements.

Definition B.1 (Harmonic Measure)**.** *The harmonic measure $\omega(z, E, \Omega)$ is the probability that a Brownian motion starting at $z \in \Omega$ first hits the boundary $\partial\Omega$ in the set $E \subset \partial\Omega$.*

Theorem B.1 (Kakutani's Theorem)**.** *If D is a bounded domain and g is a continuous function on ∂D, then the function defined by:*

$$u(x) = \mathbb{E}^x[g(B_{\tau_D})] = \int_{\partial D} g(y)\, d\omega_x(y) \tag{B.3}$$

is harmonic in D and satisfies $\lim_{x \to y} u(x) = g(y)$ for regular boundary points $y \in \partial D$.

B.5 References

This is an example of a citation. Witten and Sander introduced the model in 1981 [1].